USN | | | | | | | | | | [24ESEE11]

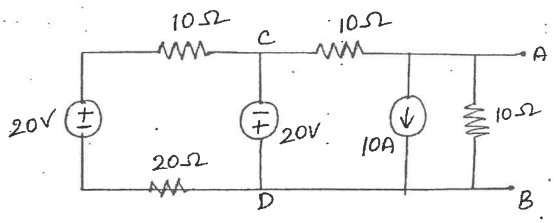
FIRST SEMESTER B.Tech EXAMINATION, JANUARY-2025

ELEMENTS OF ELECTRICAL AND ELECTRONICS ENGINEERING

Time: 3 Hours **Maximum Marks: 100**

Instructions: **i.**Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		Questions	Blooms Level	CO's	Marks
		PART-A	**5 x2=10**		
1.	**a.**	Explain the steps associated with nodal analysis.	L2	CO 1	**02**
	b.	State Biot-Savart's Law	L2	CO 2	**02**
	c.	Define Apparent power.	L2	CO 3	**02**
	d.	Write any four applications of Nano Resistors.	L2	CO 4	**02**
	e.	Give the expansion for ELCB, RCCB and MCB	L2	CO 6	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Reduce the given circuit using Source transformation technique into a single voltage source connected in series with a resistor. 	L3	CO 1	**05**
		OR			
	b.	Give the colour codes used to identify the value of a resistor hence explain the effect of temperature on Resistor.	L2	CO 1	**05**
3.	**a.**	A probe of the magnetometer is moved around an area of 0.6 m^2 near a large sheet of magnetic material and indicates a constant reading of 5 mT. calculate the magnetic flux through that area. What has to be done if the magnetic field reading changes over the above said area?	L3	CO 2	**05**
		OR			
	b.	Differentiate between Permeance and Reluctance with their expression	L2	CO 2	**05**
4.	**a.**	Load resistors, R_1, R_2, and R_3 are each 100 Ω and connected in star with a phase voltage V_p=100 V. Determine: (a) the line current, (b) the neutral current, and (c) the line voltage.	L3	CO 3	**05**

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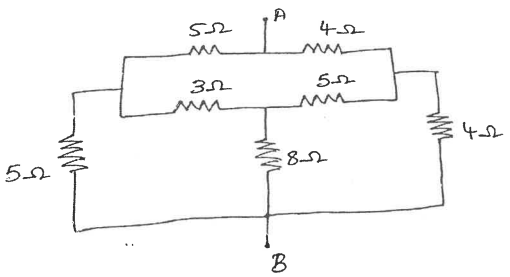
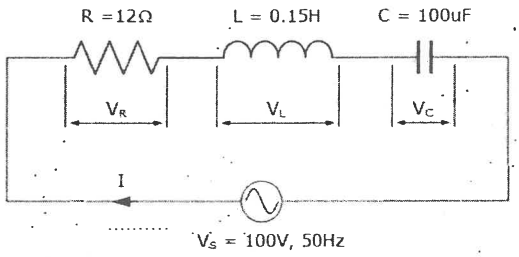
		OR			
	b.	In a system of three phase 4-wire star systems, with R1=50 Ohms, R2=30 Ohms, R3= 20 Ohms calculate the calculate the current and voltage if the given line voltage is 440 V.	L3	CO 3	**05**
5.	**a.**	Explain the color coding of resistor with an example for 3 band and 4 band resistor coding system.	L2	CO 4	**05**
		OR			
	b.	Explain the different types of materials used for Nano capacitors.	L2	CO 4	**05**
6.	**a.**	Discuss on various types of wiring system that can be used in a residential accommodation	L2	CO 5	**05**
		OR			
	b.	Explain the different types of the Electric Lamps giving their pros and cons	L2	CO 5	**05**
7.	**a.**	Explain the role of each part of a MCB with a neat structure diagram.	L2	CO 6	**05**
		OR			
	b.	If a high voltage line falls on the ground and a person is standing near to it how he can escape from getting electrocuted.	L2	CO 6	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Determine the resistance between the terminals A and B of the network shown, using STAR/DELTA Transformation. 	L3	CO 1	**10**
		OR			
	b.	Explain the VI characteristics and charging and discharging of Capacitors with time constant.	L2	CO 1	**10**
9.	**a.**	Derive the expression for the magnetic field at a point P located at a distance R from an infinitely long straight wire that carries a current I.	L3	CO 2	**10**
		OR			

	b.	What is reluctance? Give the reluctance of a magnetic bar hence determine the reluctance when the magnetic bars are connected in series	L2	CO 2	**10**
10.	**a.**	Derive the relation between line and phase voltage in a Star connection hence give the differences between a balanced and unbalanced 3 phase loads.	L2	CO 3	**10**
		OR			
	b.	A series RLC circuit containing a resistance of 12Ω, an inductance of 0.15H and a capacitor of 100uF are connected in series across a 100V, 50Hz supply. Calculate the total circuit impedance, the circuits current, power factor and draw the voltage phasor diagram. 	L3	CO 3	**10**
11.	**a.**	Explain the Evolution of electronics – Vacuum tubes to Nano electronics.	L2	CO 4	**10**
		OR			
	b.	Explain briefly the Primary cells and Secondary cells with suitable examples and give their advantages and drawbacks.	L2	CO 4	**10**
12.	**a.**	Using a neat sketch, describe in detail pipe Earthing, give its merits and drawbacks. Also mention how a pipe Earthing has to be maintained?	L2	CO 5	**10**
		OR			
	b.	Compare various types of electric lamps by considering various parameters including cost, life etc. that are used in a domestic installation and give your recommendations based on the comparison.	L2	CO 5	**10**
13.	**a.**	Discuss in detail the various safety rules that has to be followed in an electric installation by mentioning their standard as laid down by the Govt. of India	L2	CO 6	**10**
		OR			
	b.	Explain the digital multi-meter and give the role of each part with a block diagram.	L2	CO 6	**10**